

Retail Sales Analysis SQL Project

Comprehensive Analysis Report

Executive Summary

This retail sales analysis examines transaction data across major categories and highlights insights on sales performance, customer behavior, and operational patterns.

1. Project Scope & Methodology

- Data Source: retail_db
- Time Period: Multi-year
- Sample Size: 10,000+ transactions
- Unique Customers: 1,500+
- Approach: SQL-driven EDA and diagnostic analysis

2. Data Quality & Preparation

Initial checks indicated missing values which were removed for integrity.

3. Key Insights

Sales Performance:

- Category-level revenue dispersion.
- High-value transactions > \$1,000 found.

Customer Behavior:

- Top 5 customers contribute major revenue.
- Clear demographic patterns by category.

Temporal Patterns:

- Monthly performance ranking using window functions.
- Time-of-day impact on order count.

4. SQL Techniques Used

- Aggregations (SUM, COUNT, AVG)
- Window functions (RANK, PARTITION BY)
- Date functions (EXTRACT, TO_CHAR)

- Cleaning & validation queries

5. Business Recommendations

- Launch loyalty program
- Optimize inventory by time-of-day/month patterns
- Targeted marketing based on demographic insights

6. Limitations

- Lacks external factors (holidays, weather)
- No LTV or customer journey data

7. Future Directions

- Time-series forecasting
- Customer clustering
- Real-time analytics

Prepared By: Data Analytics Team